ENGINEERING PORTFOLIO

Zhongxiao (John) Liu

Mechanical Design / Robotics Hardware / Systems Integration

Selected projects spanning fixture concepts, finite-element analysis, mechanical CAD, physical assembly, and ROS 2 hardware integration.



On-site assembly of the safety-valve test platform.

SELECTED PROJECTS

01	**Safety valve test platform**	Completed
02	**ROS 2 mobile robot**	Completed
03	**U-bend gripper optimization**	In Progress
04	**SafeStep**	In Progress

zhongxiao.liu@duke.edu | github.com/liu-cool-cmd

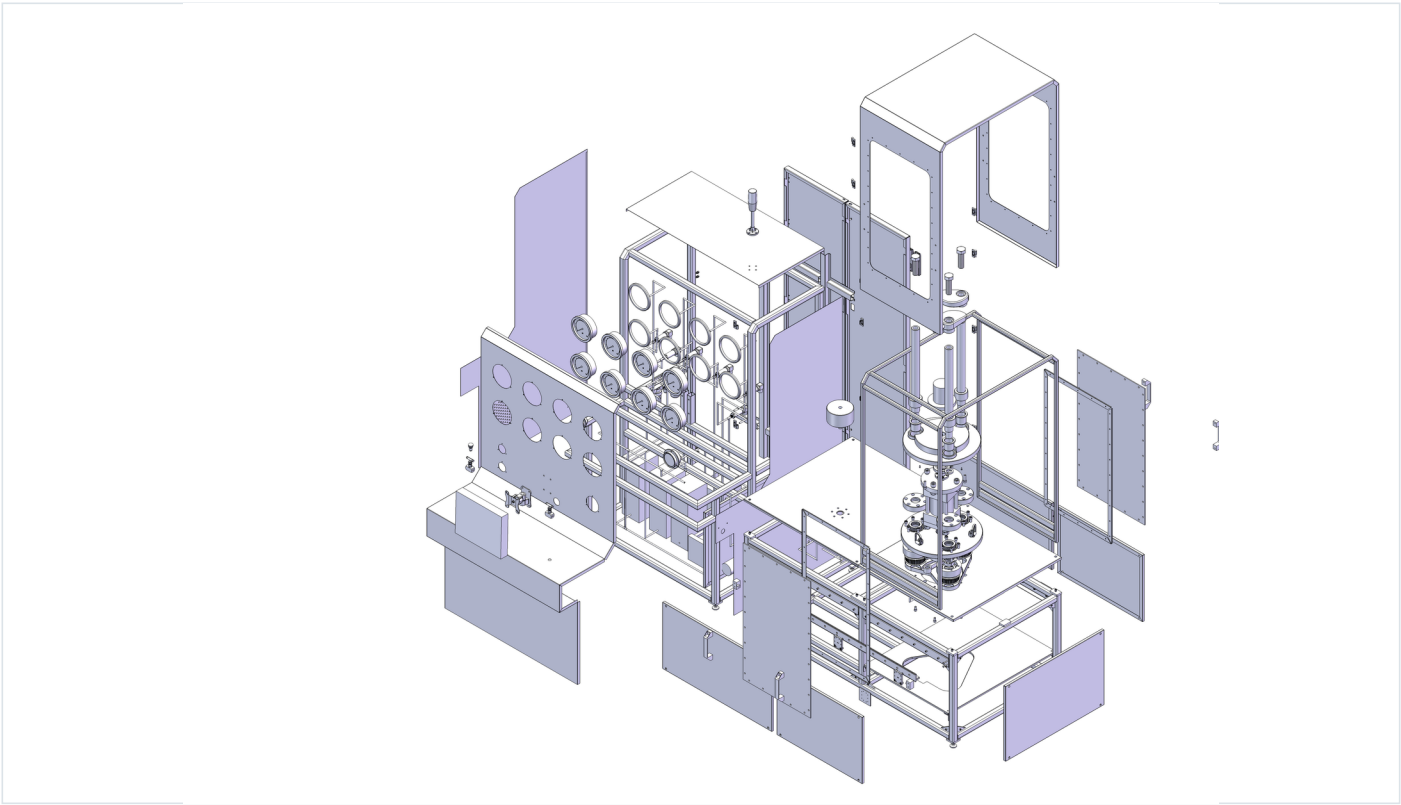
01 / COMPLETED CAPSTONE

Safety valve test platform

Completed

Donghua University / Sep 2025 - May 2026

The platform holds safety valves for loading, positioning, and access to the test area. My work covered fixture concepts and FEA, mechanical modeling, fabrication deliverables, and on-site assembly.



Exploded assembly: sheet-metal panels, rectangular-tube frames, fixture, service doors, and sliding enclosure.

MY CONTRIBUTION

Fixture concepts + FEA
Compared clamping concepts and refined contact geometry.

Sheet-metal modeling
Folded panels, openings, reinforcement, and fastening details.

Rectangular-tube frames
Control-console and fixture frames with mounting interfaces.

Overall mechanical assembly
Integrated frames, panels, fixture, and enclosure in CAD.

Sliding-door system
Moving frame, rail mounts, windows, and retaining hardware.

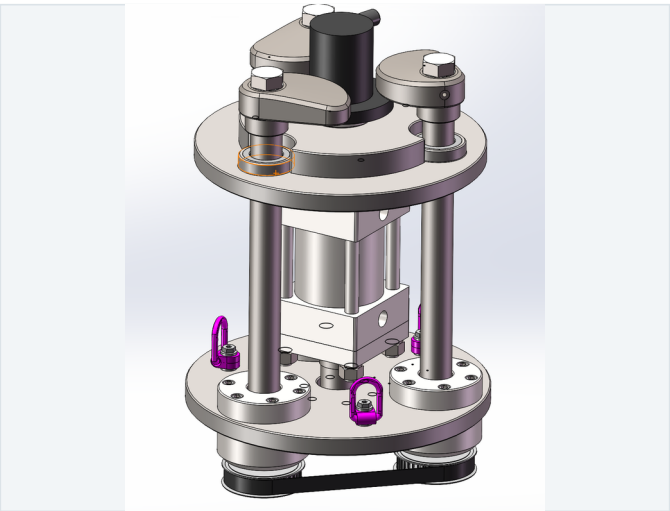
Fabrication + assembly
Mechanical BOM and STEP files; on-site fit troubleshooting.

01 / FIXTURE CONCEPTS + FEA

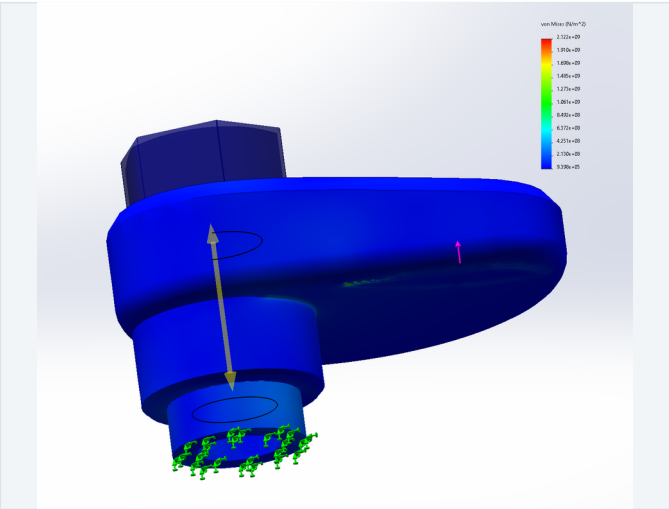
Clamping and contact decisions

The fixture considered DN15-DN100 valve flanges. Three eccentric elements provide radial positioning; hydraulic loading supplies axial force. A timing belt coordinates the eccentric elements.

FEA setup: one-third symmetric model; locally refined contact mesh; static contact analysis with large-displacement behavior.

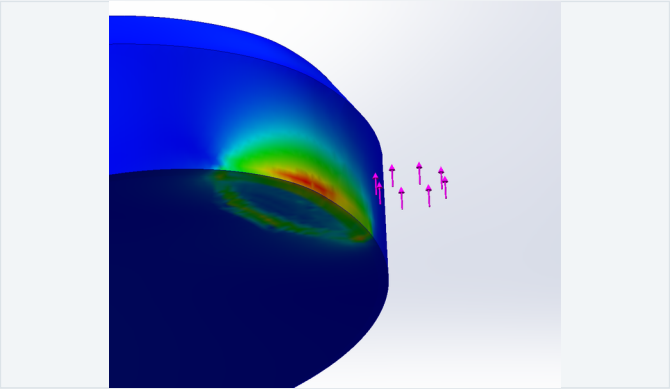


CAD: the fixture assembly and mechanical load path.

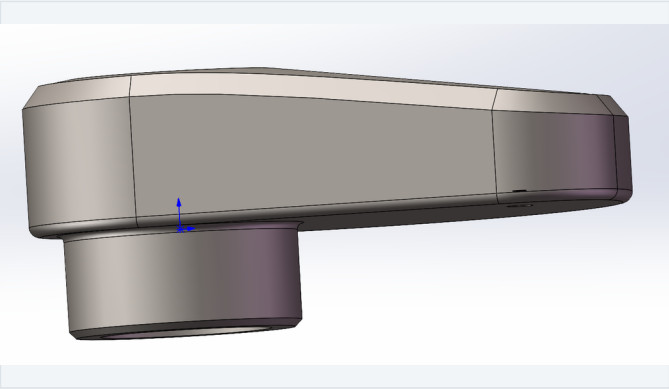


Simulation: stress plot used to review the clamping load path.

Local contact geometry guided the revision



Simulation: concentrated loading near the jaw-flange interface.



CAD: R10 mm transition at the contact edge.

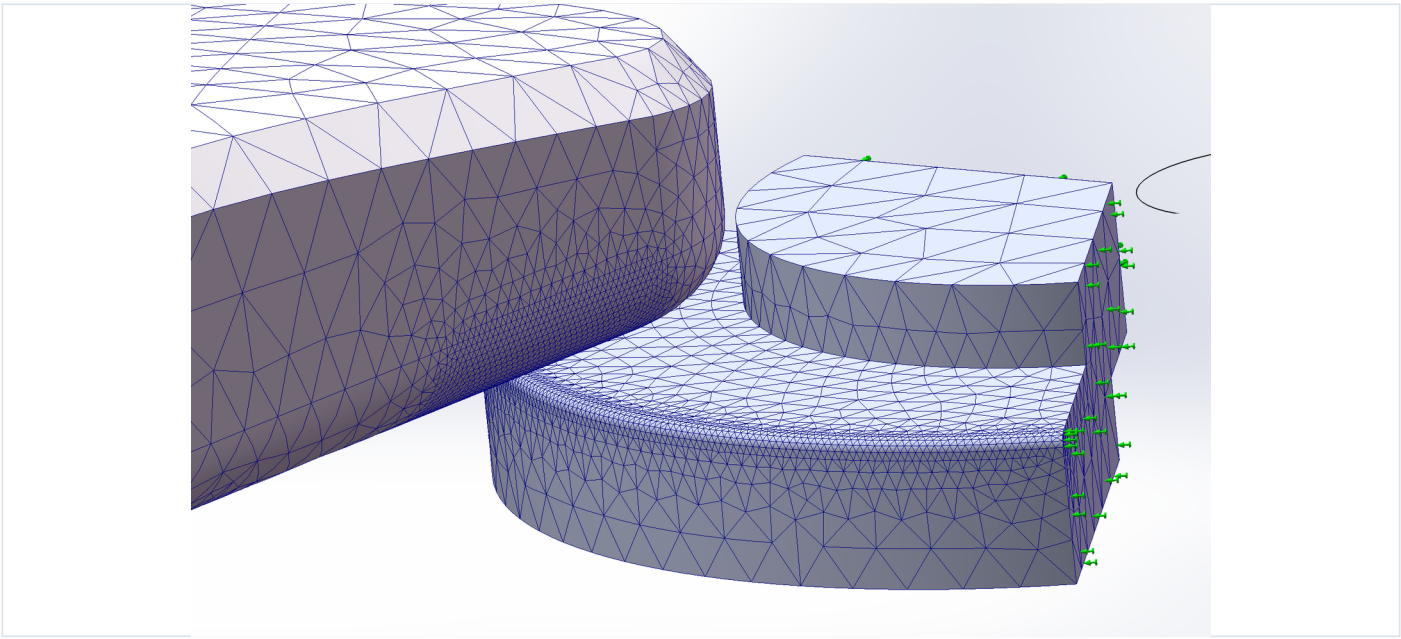
Design decision

FEA showed localized stress concentration at the jaw-flange interface. I increased the transition radius at the contact edge to address it.

01 / ANALYSIS METHOD + DESIGN DETAILS

Contact analysis and mounting details

I set up the SolidWorks Simulation study around the repeated jaw-flange interface, concentrating model resolution at the contact region.



Analysis setup: locally refined jaw-flange mesh and applied constraints.

1 / Symmetry and contact

One 120-degree sector with symmetry fixtures at the cut faces; local no-penetration jaw-flange contact with friction coefficient 0.15.

2 / Mesh and solution

Curvature-based, second-order tetrahedra; 1 mm contact-region mesh control; linear-elastic materials; large-displacement static study with the Direct Sparse solver.

Threaded supports in hollow tube

At the pump mounts, 18 mm diameter x 30 mm inserts with M10 threads transfer fastening loads into solid supports inside the tube. Rubber isolators connect the pump feet to these mounts.

Window and rail interfaces

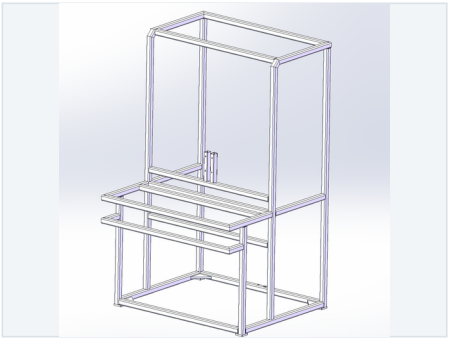
Two 5 mm polycarbonate panels are held by retaining strips and M6 x 20 self-clinching fasteners. Four welded plates provide rail mounting interfaces on the moving frame.

Manufacturing clearance: the manufacturing team added 0.5-1.5 mm seam allowances for weld distortion and bend springback, then welded and ground the joints.

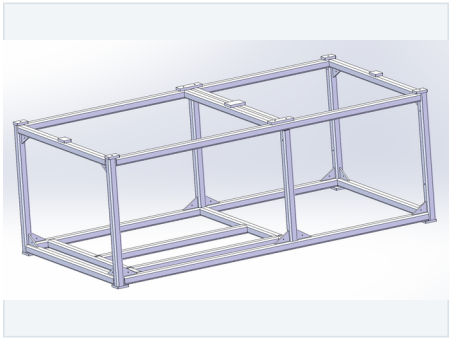
01 / MECHANICAL CAD + FABRICATION

Frames, panels, and sliding enclosure

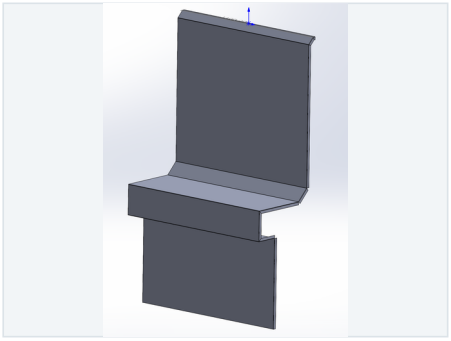
I modeled 80 x 40 x 4 mm tube frames and a 2 mm sheet-metal enclosure. Flush rivet nuts support panel fit; rear-door reinforcement and welded M6 x 18 studs integrate the lock hardware.



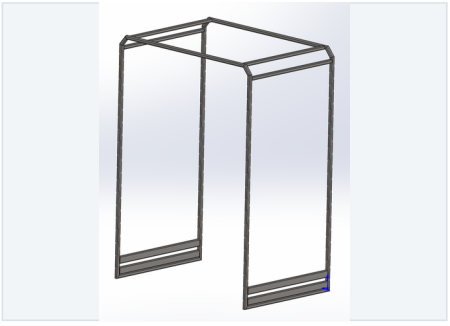
Console tube frame



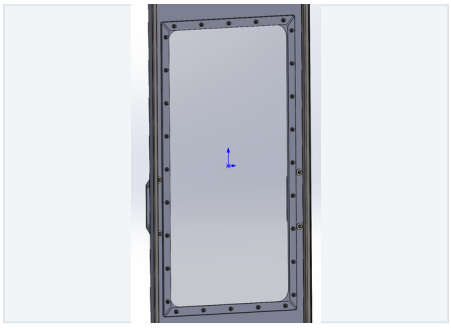
Fixture-side tube frame



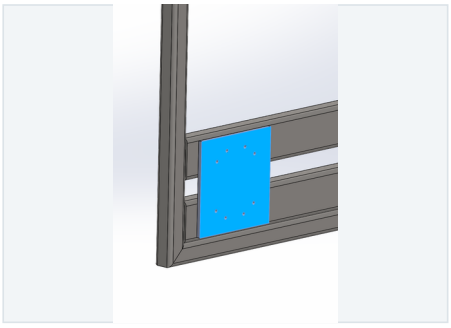
Folded sheet-metal panel



Sliding enclosure frame



5 mm PC panels / M6 x 20 fasteners



Four welded rail mounting plates

On-site integration: rail-frame interference

During assembly, I identified interference at the rail-frame interface and performed on-site mechanical rework.



Fabricated frame and enclosure.



Rail interface during assembly.

01 / BUILT HARDWARE

Completed mechanical assembly

The assembled platform integrates the sheet-metal console, rectangular-tube frames, valve fixture, and sliding enclosure.



Workshop view of the assembled console and fixture-side structure.



Front view showing the fixture, sliding enclosure, and mechanical interfaces.

02 / COMPLETED INTEGRATION PROJECT

ROS 2 mobile robot

Completed - Bring-up + Mapping

Jan - Aug 2026

I integrated a commercial Mecanum-wheel chassis, Jetson computer, LiDAR, and depth camera for ROS 2 teleoperation and indoor mapping.

COMPUTE
Jetson Orin Nano 8 GB

MOBILE BASE
Yahboom X3 / STM32 controller

DEPTH CAMERA
Orbbec Astra Pro

SOFTWARE
Ubuntu 22.04 / ROS 2 Humble

LIDAR
2D TOF LiDAR

GROUND STATION
Native Ubuntu laptop / RViz

Hardware interfaces

I built and configured the workspaces, established stable device aliases, and checked chassis feedback, LiDAR scans, camera topics, and the robot model.

Remote inspection

The robot and Ubuntu ground station shared a network and ROS domain. I used topic inspection and RViz to diagnose the interfaces and observe system output.

Selected troubleshooting

01 / Odometry sign mismatch

Symptom: forward physical motion appeared as negative velocity in `/vel_raw`.

Change: I traced the mismatch to driver publication logic and corrected the signs before publishing Twist.

Check: after restarting the driver, forward motion produced positive forward odometry.

02 / Serial devices changed names

Symptom: chassis and LiDAR paths depended on USB enumeration order.

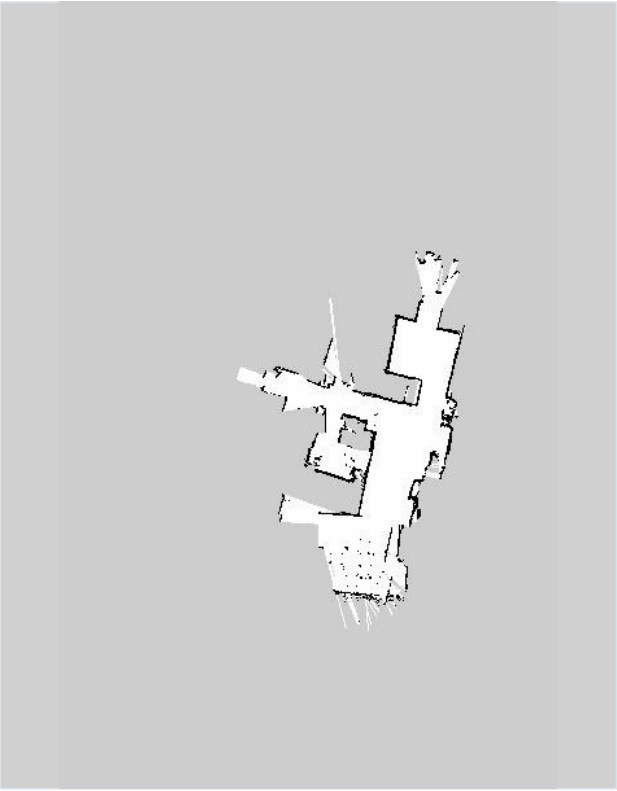
Change: I configured udev aliases using vendor/product IDs and addressed the chassis serial-driver and service conflict.

Check: I verified the devices and used the persistent paths in driver configuration.

02 / CALIBRATION + MAPPING

From sensor interfaces to indoor maps

I first used Gmapping to exercise the motion-odometry-LiDAR-TF chain. I also tested Cartographer after observing odometry drift and explored loop closure when revisiting mapped areas.



Indoor occupancy map generated during mapping tests.

Motion calibration

I used the vendor linear and angular calibration nodes to compare commanded motion with physical distance and rotation, then updated the scale factors.

I inspected maps in RViz and saved them using the Nav2 map server.

03 / Time synchronization

Jetson clock drift caused TF timing issues and clock-skew warnings. I configured Chrony and an initial clock correction policy to address them.

04 / Ground-station environment

WSL2 network discovery was unreliable for the workflow. I switched to a native Ubuntu laptop with the same ROS domain and used it for topic inspection and RViz.

03 / DUKE UNIVERSITY + TRANE TECHNOLOGIES

U-bend gripper optimization

In Progress

Sep 2026 - Present

This Duke-Trane course project focuses on improving insertion accuracy in an existing factory system for robotic U-bend insertion during evaporator-coil assembly. My work focuses on gripper optimization for consistent part positioning, insertion alignment, and copper protection.



Coil end face with copper U-bends.

Copyright Trane | SureFit Coils

Uneven gripping can tilt the U-bend, causing one leg to contact a tube rim before the other and increasing the risk of jamming or copper damage.

Gripper design priorities

- Consistent U-bend positioning and tilt control.
- Clearance for gripper opening and withdrawal near neighboring tubes and corners.
- Stable gripping with protection against copper damage.
- Manufacturable and replaceable contact parts.

04 / WEARABLE SENSING COURSE PROJECT

SafeStep

In Progress

Duke University / Sep 2026 - Present

PROJECT BACKGROUND

SafeStep is a course project exploring a low-cost sensing insole for partial weight-bearing rehabilitation after lower-extremity surgery. The goal is to estimate affected-foot loading during walking and provide audio feedback when loading falls outside a clinician-prescribed range.

The team concept combines pressure sensing, load estimation, audio feedback, and loading-history review. Accuracy, repeatability, comfort, power consumption, and cost are linked design constraints.

MY ROLE

Team lead - sensor integration and hardware bring-up

I lead the team and am responsible for sensor integration, procurement and BOM preparation, hardware bring-up, calibration, and data streaming for the team's load-estimation work.

SENSOR SELECTION AND PROTOTYPE PLANNING

I reviewed resistive, capacitive, and strain-gauge sensor options and worked on requirements for an eight-point prototype. The prototype plan pairs a ZNX-01 sensor insole with ESP32 acquisition hardware and an initial BOM.

Hardware tradeoffs

Sensor density

Eight sensing points simplify wiring and processing while reducing spatial detail.

Electronics placement

Internal packaging adds shoe thickness; external hardware adds wiring.

Cost and sensing

Calibration and repeatability are central to estimating force with low-cost sensors.